

Karthikeya Machiraju

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Research Interests

I work in quantum computational physics – using quantum simulation, digital and analog, to study many-body and quantum-field-theory problems that are hard to reach classically. My current work spans variational quantum simulation of lattice gauge theories (the Schwinger model), non-equilibrium dynamics in periodically driven spin systems, and entanglement in open photonic platforms. Looking ahead, I want to work on real-time and finite-density simulation of gauge theories – confinement, string breaking, and thermalization – using both near-term quantum hardware and tensor-network methods.

Education

- PES University, Bengaluru** – B.Tech, Computer Science (AI/ML) 2023–2027 (expected)
- Physics-relevant coursework: Quantum Entanglement and Quantum Computing; Quantum Transport and Logic Gates; Nonlinear Optics and Quantum Technology; Linear Algebra and its Applications; Design and Analysis of Algorithms.
- Indian Institute of Technology Madras (Distance Learning)** – B.Sc, Data Science 2023–Present
- Statistics for Data Science I–II; Mathematics for Data Science I–II; Computational Thinking; Programming in Python.

Publications & Manuscripts

SUBMITTED = *under review*; READY = *manuscript complete*; DRAFTING = *in active write-up*.

- [1] K. Machiraju, K. Sujith, K. Bhowmick. *Variational Quantum Simulation of the Schwinger Model*. READY.
- Implemented the Schwinger model at circuit level using VQE (**Cirq**), reproducing trapped-ion benchmarks to within 0.013% and analysing how noise, ansatz expressibility, and barren plateaus limit near-term lattice-gauge simulation.
 - Introduced von Neumann entanglement entropy, via Schmidt decomposition, as a noise-robust order parameter for the first-order phase transition – more stable than energy-based diagnostics under hardware noise. [**Cirq**, VQE, exact diagonalisation]
- [2] K. Machiraju, K. Bhowmick. *Drive-Structure vs. Geometry: Two Distinct Protection Classes in Floquet and Rondeau Time Crystals*. DRAFTING.
- Identified two distinct prethermal protection mechanisms in driven spin systems with opposite responses to dimensionality, correcting a $19.2\times$ mean-field estimate to a measured $5.0\times$ via exact diagonalisation.
 - The drive-structural (rondeau) protocol matches 2D Floquet time-crystal stability on a plain 1D device with better noise robustness, via a two-line change to the classical drive table. [**Cirq**, **QuTiP**, exact diagonalisation up to $N = 12$]
- [3] K. Machiraju, K. Venkatesh, G. Rao M, N. P. Shetty, G. Srinivasa. *Revisiting Quantum Generative Models for Imbalanced Learning: Calibration-Aware Evaluation and Dimensional Effects*. SUBMITTED, IEEE QCE 2026.
- Benchmarked quantum generative models against classical baselines on high-energy physics datasets, identifying a dimension-dependent regime where quantum models lead classical ones by up to +5.8 pp F1.
 - The gap closes under dimensionality reduction, clarifying when the advantage holds and when it does not for this task; hardware runs on IBM Fez characterised a $2.2\times$ entropy increase under NISQ noise. [**PennyLane**, **Qiskit**, IBM Fez hardware]
- [4] A. K. Dwivedi, K. Machiraju, K. Bhowmick. *Robustness Against Disorder of Holonomic Quantum Gates in Photonic Waveguide Architectures*. SUBMITTED, Quantum Information Processing (Springer).
- Showed that holonomic gates are robust to control disorder but not to environmental decoherence – a separation standard fidelity metrics do not capture: fidelity stays ≈ 0.997 while concurrence falls to 0.49 under $1/f$ noise.
 - Found that combining holonomic control with dynamical decoupling mitigates both error types, with the pulse-number response distinguishing the dominant decoherence channel. [**QuTiP**, Lindblad open-system simulation]
- [5] K. Sujith, K. Machiraju, K. Bhowmick. *Closed-Form Interference Optimum and Parametric Realization of Photon Blockade in Weakly Nonlinear Si_3N_4 Cavities*. SUBMITTED, FiO+LS 2026.
- Derived closed-form unconventional photon-blockade conditions giving $g^{(2)}(0) = 4.3 \times 10^{-5}$ at $U/\kappa = 0.1$, and mapped a parametric Si_3N_4 realisation reaching $g^{(2)}(0) = 0.28$ at 100% yield under $\pm 5\%$ geometric variance. [analytic optimum + parametric device modelling]

[6] K. Sujith, K. Machiraju, K. Bhowmick. *Bulk Optical Loss Mechanisms and Dynamic Signal Distortion in Polymer and Glass Photonic Crystal Waveguides for Co-Packaged Optics*. SUBMITTED, FiO+LS 2026.

- Built a five-channel loss taxonomy and closed-form dynamic loss model (Beer–Lambert plus thermo-optic detuning) for photonic-crystal waveguide links; 50 Gb/s simulations at +20 dBm show polymer links fail across the practical range of group indices, versus error-free glass operation. [link-level signal-integrity simulation]

Research Experience

Undergraduate Researcher – Photonics and Quantum Technology Lab, PES University

Jan 2025–Present

Advisor: Dr. Kaustav Bhowmick, Associate Professor, Department of Electronics & Communication

Digital quantum simulation of the Schwinger model. Built a full circuit-level implementation of the two-flavor Schwinger model using variational quantum eigensolver methods (**Cirq**), reproducing trapped-ion experimental benchmarks to within 0.013%. Analysed noise effects, ansatz expressibility, and the barren-plateau problem to understand what limits near-term lattice-gauge simulation; introduced von Neumann entanglement entropy via Schmidt decomposition as a noise-robust order parameter for the first-order phase transition. [Paper 1.]

Non-equilibrium dynamics in Floquet spin systems. Studied thermalization and prethermal protection in periodically driven spin chains and 2D lattices using density-matrix simulations and exact diagonalisation (**Cirq**, **QuTiP**). Found that geometry and drive structure protect time-crystal order through distinct mechanisms; this became clear only after correcting an initial mean-field estimate of $19.2\times$ against exact numerics (measured $5.0\times$). [Paper 2.]

Decoherence and entanglement in open photonic systems. Investigated coherence-extension strategies for Bell-pair sources under realistic photonic dephasing by coupling auxiliary intraparticle entanglement degrees of freedom (**QuTiP**, Lindblad/Kraus); exhaustive open-system simulation isolated an entanglement-monogamy obstruction that bounds the achievable gain. Separately showed that standard gate-fidelity metrics do not detect decoherence-induced entanglement loss in two-qubit gates – fidelity stays high while concurrence falls. [Paper 4.]

Teaching & Academic Service

Teaching Assistant – Quantum Entanglement and Quantum Computing, PES University

Aug–Dec 2025

- Led lab sessions on quantum algorithms, entanglement measures, and quantum channels; lectured on open quantum systems and Kraus representations; mentored student projects on variational circuits and noise simulation on IBM Quantum hardware.

Teaching Assistant – Quantum Transport and Logic Gates, PES University

Jan 2026–Present

- Conducting lab sessions on quantum transport, Lindblad dissipation, and hardware-aware gate realisations; lectured on loss mechanisms and gate robustness; supporting projects on noise-aware circuit design.

Subject Matter Expert – Introduction to Quantum Computing, PESU IO

Aug–Nov 2025

- Designed and delivered a one-month undergraduate course on quantum gates, VQE, QAOA, and noise-aware circuits with IBM Quantum simulation labs.

Leadership

Club Head – Q-Forest Quantum Computing Society, PES University

May 2024–Present

- Lead a 100+-member society; organise research bootcamps, arXiv reading groups, and annual symposia (*Quanta*, *Graviton 2.0*).

Research Head – Equinox Space Technology Society, PES University

May 2024–Present

- Coordinate research in computational astrophysics, quantum sensing, and space-oriented quantum technologies; mentor undergraduate simulation projects.

Technical Skills

Quantum: Cirq, Qiskit, QuTiP, PennyLane; VQE/QAOA; exact diagonalisation ($N \leq 12$); Lindblad/Kraus/Choi open-system methods; Floquet simulation; lattice gauge theory (Jordan–Wigner, Kogut–Susskind).

Scientific computing: Python (NumPy, SciPy, SymPy), MATLAB, CUDA, Git, HPC workflows; PyTorch, scikit-learn.

Honors & Additional Training

IBM Qiskit Advocate (2025) · Womanium Quantum + AI Program (2024) · Introduction to Quantum Information Science, KAIST.